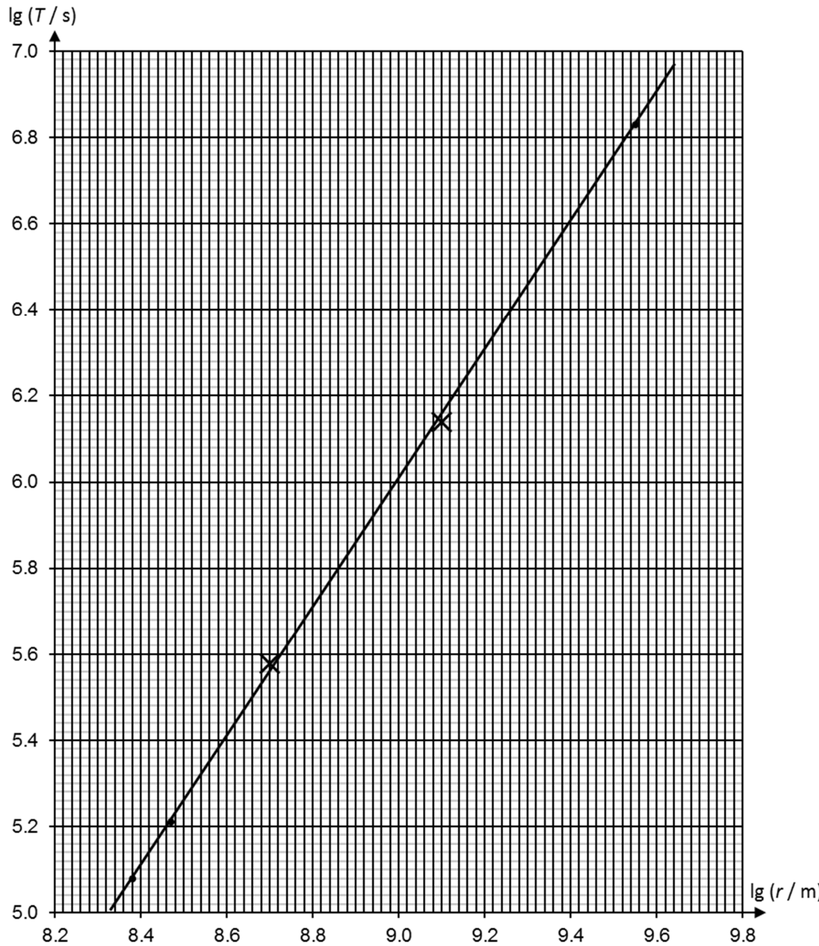
7 (a)	Let the mass of the satellite be *m*. According to Newton's Law of Gravitation, $F_G = \frac{GMm}{r^2}$ Centripetal force $F_c = mr\omega^2$	M1
	When the satellite orbits around the planet in a circular orbit, the gravitational force acting on the satellite by the planet **provides** the centripetal force for the orbit. $F_C = F_G$ $mr\omega^2 = \frac{GMm}{r^2}$	M1
	$mr(\frac{2\pi}{T})^2 = \frac{GMm}{r^2}$ $T^2 = \frac{4\pi^2}{GM} r^3$	M1 A0

(b)	Rhea: $\lg T = \lg(0.380 \times 10^6) = 5.58$ $\lg r = \lg(0.501 \times 10^9) = 8.70$	B1
	Titan: $\lg T = \lg(1.38 \times 10^6) = 6.14$ $\lg r = \lg(1.26 \times 10^9) = 9.10$	B1
(c)(i)	 correct plots	B1
(c)(ii)	Best fit line	B1
(d)(i)	Gradient $= \dfrac{6.60 - 5.11}{9.40 - 8.40}$	M1
	= 1.49 (within 1.47 – 1.53)	A1

(d)(ii)	The moons of Saturn which orbit in circular paths must obey $T^2 = \frac{4\pi^2}{GM}r^3$. Linearising this equation will lead to the equation $\lg T = \frac{1}{2}\lg\frac{4\pi^2}{GM} + \frac{3}{2}\lg r$.	M1
	On a graph showing the variation of $\lg T$ with $\lg r$, this equation is represented by a straight line with a **gradient of 1.5**. Since the gradient of the graph in Fig 7.2 is 1.49, the data supports he relationship in **(a)**.	A1
(e)	$\lg(3.78 \times 10^8) = 8.58$ From the graph, the corresponding value for $\lg T$ is 5.38.	C1
	Hence, $T = 10^{5.38} = 2.40 \times 10^5$ s	A1
(f)(i)	$$\lg T = \frac{1}{2}\lg\frac{4\pi^2}{GM} + (1.49)\lg r$$ Substituting (9.40, 6.60) $$6.60 = \frac{1}{2}\lg\frac{4\pi^2}{(6.67 \times 10^{-11})M} + (1.49)(9.40)$$	M1
	$M = 3.84 \times 10^{26}$ kg	A1
(f)(ii)	Multiple sets of readings were used in calculating the mass in **(f)(i)**, as compared to only one set used in the student's calculation. Using multiple sets of readings help to **reduce the random error** through the use of a best fit line.	B1
	This means that the student's calculation will be less accurate than the values calculated in **(f)(i)**.	B1
(f)(iii)	Using $T = 7.78 \times 10^4$ s, $$\lg(7.78 \times 10^4) = \frac{1}{2}\lg\frac{4\pi^2}{(6.67 \times 10^{-11})(3.84 \times 10^{26})} + 1.49\lg r$$ r = 1.79 ×10^5 km	M1
	r is the orbital radius, which is the distance from the centre of Saturn to the centre of the Moon Mimas, not from the surface of Saturn.	M1
	Thus the report is inaccurate.	A1